IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

EEE/EIE PART III/IV: MEng, BEng and ACGI:

MACHINE REASONING

Monday, 4 December 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : C. Qin

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ELEC60004/70068. MACHINE REASONING

1. This question is about search space formalisation and the General Graph Search Engine (GGSE), and 'uninformed' search algorithms.

   The Amazing Word Machine (AMW) has a set of rules which transform one word into another. It uses the following rules:

   1. Add any vowel to the start of a word, i.e. $MAZE \rightarrow AMAZE$.
   2. Add any consonant to the end of a word, i.e. $CARD \rightarrow CARDS$.
   3. Delete two consecutive vowels, i.e. $PLEASE \rightarrow PLSE$.
   4. Delete two consecutive consonants, i.e. $STRING \rightarrow RING$, $TRING$ or $STRI$.
   5. Double the entire word. i.e. $AYE \rightarrow AYEAYE$.

   Starting with *LEAD*, you are asked to find a way to make *GOLD*.

   a) You are given two predicates `consonant/1` and `vowel/1`. What would be the expected difference between the queries:

   ```
   ?- vowel( X ).
   ```

   and

   ```
   ?- vowel( e ).
   ```

   [4]

   b) Using these two given predicates, formulate the state space so that it could be used with the GGSE:

   (*i*) Specify a state representation;
   (*ii*) Specify the initial state;
   (*iii*) Specify the final state; and
   (*iv*) Specify the state-change rules.

   [8]

   c) Briefly explain the difference between *breadth first*, *depth first*, *uniform cost* and *iterative deepening depth first* search algorithms.

   Using appropriate evaluation criteria, compare and contrast the four algorithms.

   Briefly comment on the suitability of each algorithm for solving word transformation problems with the AMW.

   [6]

   d) If Rule 3 and and Rule 4 were changed to delete *two or more* consecutive vowels or consonants (respectively), explain (declaratively) how to generate all solutions.

   [2]

2. This question is about search space formalisation and the General Graph Search Engine (GGSE), and 'informed' search algorithms.

Two adults and two children need to cross a river in a rowing boat, subject to the following conditions:

1. All adults weigh the same.
2. Each child weighs half as much as an adult.
3. The boat can only carry the weight equivalent of one adult.
4. Adults and children can row the boat.
5. The boat must have someone in it to row it.

The problem is to get everyone across the river, given that they all start on the left bank (with the boat).

a) Specify a state representation, and justify your state representation with respect to alternative representations for river-crossing problems of this kind.

[4]

b) Given this state representation, formulate the state space so that it could be used with the GGSE:

(*i*) Specify the initial state;
(*ii*) Specify the final state; and
(*iii*) Specify the state-change rules.

[6]

c) With a definition of a graph $G = \langle N, E, R \rangle$ and an identified node $S \in N$, give an inductive definition for the set of paths P_{G_S} rooted on S.

[3]

d) With a definition of a graph $G' = \langle S, Op \rangle$, give an inductive definition for $P_{G'}$, the set of paths defined by G'.

[3]

e) Explain how to show that $P_{G_S} = P_{G'}$.

[2]

f) Explain how the paths in the inductive definition(s) are searched by the A* search algorithm.

[2]

3. This question is about knowledge representation and reasoning with Predicate Logic, Prolog, and **KE** for Predicate Logic.

Important: any answer to any part that doesn't use line numbers and rule references to indicate how new formulas have been derived will not get any marks.

a) Consider the following English statements:

If anyone chooses rock and anyone chooses scissors, then whoever chose rock beats whoever chose scissors.
If anyone chooses scissors and anyone chooses paper, then whoever chose scissors beats whoever chose paper.
If anyone chooses paper and anyone chooses rock, then whoever chose paper beats whoever chose rock.
The 'beats' relation is transitive.
Alana chooses rock.
Beatrice chooses scissors.
Charlotte chooses paper.

Express these statements in first-order predicate logic, and convert them to a set of Horn clauses.

[4]

b) Prove, using resolution and proof by negation, that *Alana beats Charlotte*. The proof should show all the unifiers.

[4]

c) Hence, or otherwise, express the Horn Clauses of part (a) as a Prolog program, and briefly explain whether or not the query `?- beats( alana, alana ).` succeeds.

[4]

d) Prove the following argument using the **KE** calculus for Predicate Logic:

No mammals are frogs.
All elephants are mammals.
Therefore, no elephants are frogs.

[4]

e) Using **KE** calculus for Predicate Logic, prove that the following argument is fallacious:

Some trees are tall things.
Some tall things are buildings.
Therefore, some trees are buildings.

[4]

4. This question is about reasoning in Propositional Logic and Propositional Modal Logic, using the **KE** calculus.

Important: any answer to any part that doesn't use line numbers and rule references to indicate how new formulas have been derived will not get any marks.

a) Prove, using the the **KE** calculus, that the following are theorems of Propositional Logic:

(*i*) $(p \rightarrow (q \rightarrow r)) \rightarrow (\neg r \rightarrow (p \rightarrow -q))$
(*ii*) $(p \rightarrow q) \rightarrow ((r \rightarrow \neg q) \rightarrow (p \rightarrow \neg r))$

[4]

b) In the *Obligatio Game*, a finite number n of rounds is chosen, depending on the severity of the exam. The examiner then gives the candidate successive assertions, $\phi_1, \phi_2, \ldots, \phi_n$ that she or he has to either *accept* or *reject* as each one is put forward. In the former case, ϕ_i is added to the candidate's stock of commitments; in the latter case, the negation $\neg\phi_i$ is added.

The candidate passes the exam if she maintains the consistency of her stock of commitments throughout all n assertions.

Suppose the candidate is exposed (successively) to the following three ($n = 3$) assertions:

1. $p \vee \neg r$
2. $\neg(q \wedge r)$
3. $q \leftrightarrow r$

Using the **KE** calculus, show what sequences of answers would ensure that the student passes the exam.

(Hint: use **KE** for model building, for all combinations of *accept* or *reject* of the assertions. If the proof has at least one open branch, then this implies a consistent sequence of answers.)

[8]

c) Define a Kripke Model, and explain how it is used to give a semantics for formulas of Modal Logic.

[4]

d) Explain how $\models^{\mathscr{M}}_{\alpha} \Box p$ can be true in a model $\mathscr{M}$ while $\models^{\mathscr{M}}_{\alpha} \Diamond p$ can be false in the same model.

Explain how this is not the case if the accessibility relation satisfies the serial condition.

Using the **KE** calculus for Modal Logic, prove that the characteristic formula for the serial condition, $\Box P \rightarrow \Diamond P$, is a theorem of modal logic **S5**.

[4]

5. This question is about the Simplified Event Calculus (SEC).

Consider the Glass-Filling Problem of having two glasses, one that can hold seven pints and one that can hold five pints, and the problem of getting four pints in the seven-pint glass.

a) Represent this problem in the SEC, specifying:
   (*i*) the multi-valued fluents;
   (*ii*) the `initially` facts; and
   (*iii*) `initiates` axioms for the *fill*, *empty* and either one of the *pour* (from one glass to another) events.

[12]

b) Give an example narrative, and explain how the SEC can be used to implement problem-solving by *planning* (i.e. generate a narrative (plan) that solves the problem).

[4]

c) Explain, with the use of the Glass-Filling Problem as an example, how knowledge is represented and reasoned with in the Simplified Event Calculus compared to the state-space formulation as a graph, and briefly discuss the implications for planning to solve problems of this kind.

[4]